

Midterm 2 W26

2026-03-03

Instructions

This is an in-person exam. Your exam must be complete and uploaded to Gradescope by 2:00p.

Answer the following questions and complete the exercises in RMarkdown. Please embed all of your code and push your final work to your repository. Your code must be organized, clean, and run free from errors. Remember, you must remove the `#` for any included code chunks to run. Be sure to add your name to the author header above.

Your code must knit in order to be considered. If you are stuck and cannot answer a question, then comment out your code and knit the document. You may use your notes, labs, and homework to help you complete this exam. Do not use any other resources- including AI assistance or other students' work.

Don't forget to answer any questions that are asked in the prompt! Each question must be coded; it cannot be answered by a sort in a spreadsheet or a written response only.

For all plots you create, a title and clearly labeled axes must be provided.

Be sure to push your completed midterm to your repository and upload the document to Gradescope. *Incorrectly uploading the exam to Gradescope will result in a 5 point deduction.* This exam is worth 50 points.

Load the libraries

```
library(tidyverse)
library(janitor)
library(naniar)
library(ggmap)
library(lubridate)
library(shiny)
```

API Key

Since you will build a map as part of the exam, take a moment to register your API key. You may need to look up your stadia maps (<https://client.stadiamaps.com/dashboard/overview>) API key online (please do that now).

```
register_stadiamaps("642bab98-037e-4566-b16b-c8f0b1a62925", write = FALSE)
```

About the data

This study is focused on the fasting behavior of polar bears in the Southern Beaufort (SB) and Chukchi seas (CS). The data were collected from 1983-2017 and include measurements of blood urea nitrogen (BUN) and creatinine, which are used to calculate a ratio that indicates whether a bear is fasting or not. The original paper is included in the data folder. *It may help to quickly read the abstract before starting the exam.*

The data are from: Rode, K.D., Wilson, R.R., Douglas, D.C., Muhlenbruch, V., Atwood, T.C., Regehr, E.V., Richardson, E.S., Pilfold, N.W., Derocher, A.E., Durner, G.M., Stirling, I., Amstrup, S.C., St. Martin, M., Pagano, A.M., and others, 2017, Spring fasting behavior in a marine apex predator provides an index of ecosystem productivity: *Global Change Biology*, v. 24, no. 1, p. 410–423, <https://doi.org/10.1111/gcb.13933> (<https://doi.org/10.1111/gcb.13933>).

Data resources

- <https://data.usgs.gov/datacatalog/data/USGS:ASC141>
(<https://data.usgs.gov/datacatalog/data/USGS:ASC141>)
- <https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2>
(<https://www.sciencebase.gov/catalog/item/61a1a9e7d34eb622f69743d2>)

The variables include:

- `bear_id` : A unique number given to each bear.
- `capture_date` : The data when the bear was tranquilized and assessed.
- `popn` : The population to which the bear belongs.
- `sex` : Sex of each bear.
- `age_class` : The age class of each bear (A=adult, S=subadult).
- `mating` : Whether or not the bear is pursuing a mate.
- `denning` : Whether or not the bear denned last year.
- `lat` : Latitude where the bear was located.
- `long` : Longitude where the bear was located.
- `urea_nitrogen` : Concentration of urea nitrogen.
- `creatinine` : Concentration of creatinine.

Load the data

Run the code chunk below to load the data, clean the names, and do some tidying.

```
polar_bears <-
  read_csv("data/polarBear_serumUreaCreatinine_beaufortChukchi_rode.csv") %>%
  clean_names() %>%
  select(-den_method, -no_cubs, -cub_age) %>%
  filter(age_class==c("A", "S")) %>%
  mutate(sex=toupper(sex),
         age_class=toupper(age_class))
```

1. (2 points) List the variable names and use a summary function to get an idea of the data structure.

```
names(polar_bears)
```

```
## [1] "bear_id"      "capture_date" "popn"         "sex"
## [5] "age_class"    "mating"       "denning"      "lat"
## [9] "long"        "urea_nitrogen" "creatinine"
```

```
str(polar_bears)
```

```
## tibble [519 × 11] (S3: tbl_df/tbl/data.frame)
## $ bear_id      : num [1:519] 6398 6384 6520 6559 20629 ...
## $ capture_date : Date[1:519], format: "1985-04-28" "1991-04-13" ...
## $ popn         : chr [1:519] "SB" "SB" "SB" "SB" ...
## $ sex          : chr [1:519] "F" "F" "F" "F" ...
## $ age_class    : chr [1:519] "A" "A" "A" "A" ...
## $ mating       : chr [1:519] "no" "no" "no" "no" ...
## $ denning      : chr [1:519] "yes" "yes" "yes" "yes" ...
## $ lat          : num [1:519] 70.4 70 70.9 70 70.5 ...
## $ long         : num [1:519] -148 -150 -145 -142 -145 ...
## $ urea_nitrogen: num [1:519] 12.3 3 30 18 10 ...
## $ creatinine   : num [1:519] 0.57 1.45 0.85 1.1 1 1.3 0.65 0.7 1.2 0.5 ...
```

```
glimpse(polar_bears)
```

```
## Rows: 519
## Columns: 11
## $ bear_id      <dbl> 6398, 6384, 6520, 6559, 20629, 20876, 20354, 21024, 2076...
## $ capture_date <date> 1985-04-28, 1991-04-13, 1992-04-18, 2000-04-12, 2003-04...
## $ popn         <chr> "SB", "SB", "SB", "SB", "SB", "SB", "SB", "SB", "SB", "S...
## $ sex          <chr> "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", "F", "F"...
## $ age_class    <chr> "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A", "A"...
## $ mating       <chr> "no", "no", "no", "no", "no", "no", "no", "no", "no", "no", "n...
## $ denning      <chr> "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", "yes", "...
## $ lat          <dbl> 70.3833, 70.0167, 70.9000, 69.9600, 70.4916, 71.1445, 70...
## $ long         <dbl> -147.6333, -149.5000, -145.2333, -141.8174, -144.9511, -...
## $ urea_nitrogen <dbl> 12.33, 3.00, 30.00, 18.00, 10.00, 5.00, 29.50, 29.00, 19...
## $ creatinine   <dbl> 0.57, 1.45, 0.85, 1.10, 1.00, 1.30, 0.65, 0.70, 1.20, 0...
```

2. (3 points) How are NA's represented and where are they distributed in the variables? Use the `naniar` package to visualize the distribution of NA's in the data and replace any alternative representations of NA with actual NA values.

```
polar_bears %>%
  filter(mating=="unknown"|denning=="unknown")
```

```
## # A tibble: 337 × 11
##   bear_id capture_date popn sex age_class mating denning lat long
##   <dbl> <date>      <chr> <chr> <chr>      <chr> <chr> <dbl> <dbl>
## 1    6458 1985-03-15    SB   F     A        unknown no    69.6 -139.
## 2    6754 1988-03-25    CS   F     A        unknown no    67.3 -165.
## 3    6932 1992-03-22    CS   F     A        unknown no    70.4 -162.
## 4   20335 1998-04-19    SB   F     A        unknown no    70.2 -137.
## 5    9412 2004-04-03    SB   F     A        unknown no    71.5 -156.
## 6   21052 2008-03-29    CS   F     A        unknown no    69.0 -166.
## 7   21219 2010-04-18    SB   F     A        unknown no    70.9 -143.
## 8   21184 2010-04-19    CS   F     S        unknown no    67.0 -166.
## 9   21152 2011-03-22    SB   F     A        unknown no    71.4 -156.
## 10  21512 2011-04-14    CS   F     A        unknown no    66.6 -165.
## # i 327 more rows
## # i 2 more variables: urea_nitrogen <dbl>, creatinine <dbl>
```

```
polar_bears_no_nas <- read_csv("data/polarBear_serumUreaCreatinine_beaufortChukchi_ode.
csv",na="unknown") %>%
  clean_names() %>%
  select(-den_method, -no_cubs, -cub_age) %>%
  filter(age_class==c("A", "S")) %>%
  mutate(sex=toupper(sex),
         age_class=toupper(age_class))
```

```
## Rows: 1048 Columns: 14
## — Column specification —————
## Delimiter: ","
## chr (8): Popn, Sex, AgeClass, Mating, Denning, den_method, NoCubs, CubAge
## dbl (5): BearID, Lat, Long, Urea Nitrogen, Creatinine
## date (1): Capture Date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
miss_var_summary(polar_bears_no_nas)
```

```
## # A tibble: 11 × 3
##   variable      n_miss pct_miss
##   <chr>        <int>    <num>
## 1 mating         299      57.6
## 2 denning         91      17.5
## 3 bear_id          0         0
## 4 capture_date     0         0
## 5 popn             0         0
## 6 sex              0         0
## 7 age_class        0         0
## 8 lat              0         0
## 9 long             0         0
## 10 urea_nitrogen    0         0
## 11 creatinine       0         0
```

3. (6 points) Is sampling even for both sexes? Make a count the shows the number of males and females sampled by population and make a plot to visualize the counts.

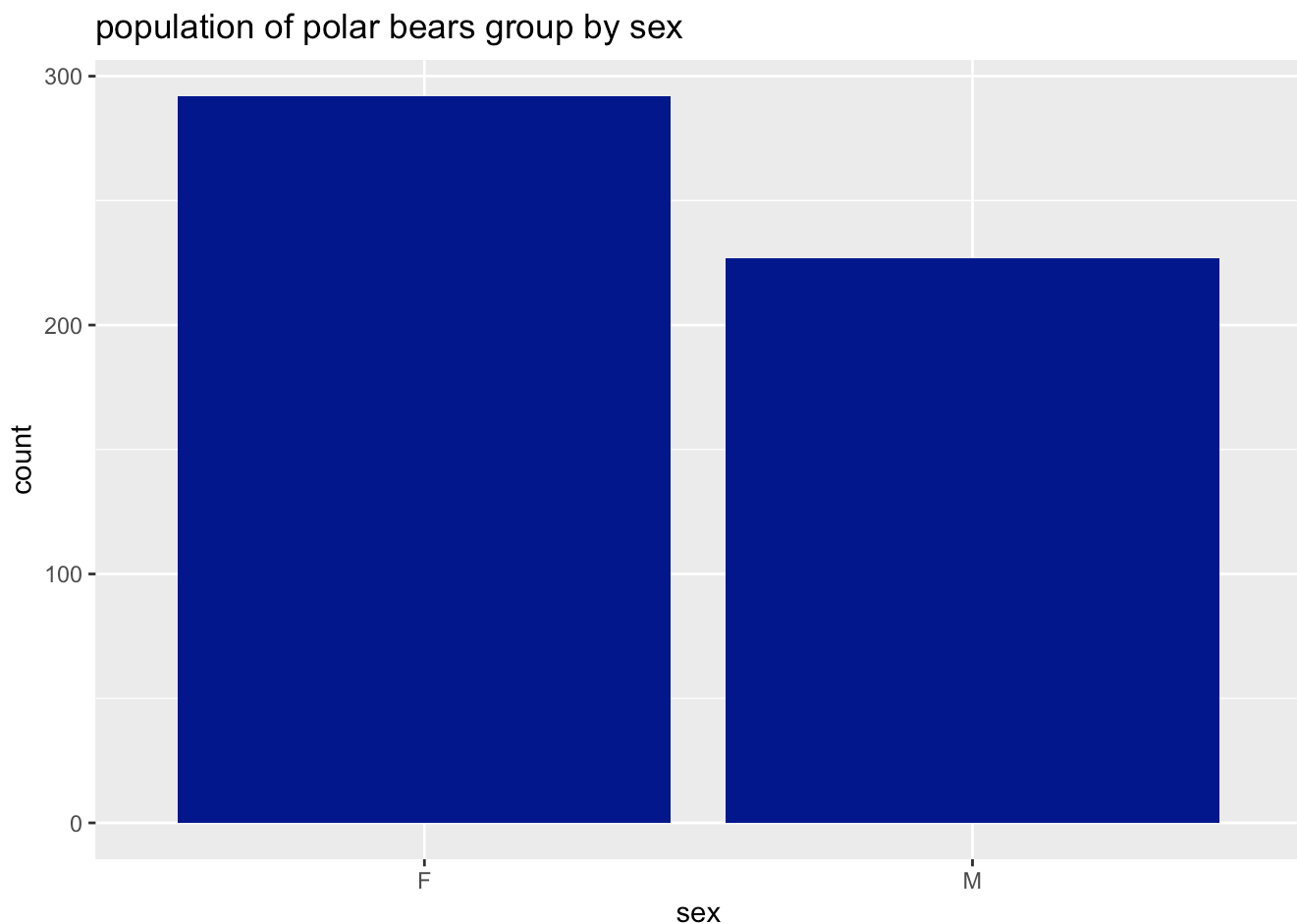
```
polar_bears %>%
  filter(sex=="F")
```

```
## # A tibble: 292 × 11
##   bear_id capture_date popn sex age_class mating denning lat long
##   <dbl> <date>      <chr> <chr> <chr>    <chr> <chr> <dbl> <dbl>
## 1   6398 1985-04-28   SB   F     A      no    yes   70.4 -148.
## 2   6384 1991-04-13   SB   F     A      no    yes   70.0 -150.
## 3   6520 1992-04-18   SB   F     A      no    yes   70.9 -145.
## 4   6559 2000-04-12   SB   F     A      no    yes   70.0 -142.
## 5  20629 2003-04-16   SB   F     A      no    yes   70.5 -145.
## 6  20876 2007-04-01   SB   F     A      no    yes   71.1 -153.
## 7  20354 2009-04-16   SB   F     A      no    yes   70.7 -145.
## 8  21024 2009-04-25   SB   F     A      no    yes   70.5 -147.
## 9  20764 2009-05-01   SB   F     A      no    yes   70.8 -149.
## 10 21045 2009-05-19   SB   F     A      no    yes   70.7 -148.
## # i 282 more rows
## # i 2 more variables: urea_nitrogen <dbl>, creatinine <dbl>
```

```
polar_bears %>%
  filter(sex=="M")
```

```
## # A tibble: 227 × 11
##   bear_id capture_date popn sex age_class mating denning lat long
##   <dbl> <date>      <chr> <chr> <chr>      <chr> <chr> <dbl> <dbl>
## 1    6096 1983-04-29    SB    M    A        yes    no    69.9 -141.
## 2    2599 1983-04-30    SB    M    A       unknown no    69.9 -141.
## 3    6312 1984-03-22    SB    M    A        yes    no    71.7 -156.
## 4    6343 1984-04-09    SB    M    S       unknown no    70.9 -147.
## 5    6358 1984-05-06    SB    M    S       unknown no    70.9 -149.
## 6    1576 1984-05-06    SB    M    A        yes    no    70.7 -148.
## 7    6374 1985-03-21    SB    M    S       unknown no    70.8 -147.
## 8    6401 1985-04-30    SB    M    A       unknown no    70.7 -147.
## 9    9974 1987-04-03    SB    M    A        yes    no    70.6 -140.
## 10   6437 1987-04-19    SB    M    S        yes    no    71.1 -146.
## # i 217 more rows
## # i 2 more variables: urea_nitrogen <dbl>, creatinine <dbl>
```

```
polar_bears %>%
  group_by(sex) %>%
  ggplot(aes(x=sex))+
  geom_bar(fill="darkblue")+
  labs(title = "population of polar bears group by sex")
```



4. (6 points) How many bears were sampled in each year for both populations? Make a line plot(s) to visualize sampling by year for both populations. (hint: you will need to pull apart the date to get the

year in its own column)

```
polar_bears_no_nas %>%
  select(capture_date) %>%
  mutate(capture_date=ymd(capture_date))
```

```
## # A tibble: 519 × 1
##   capture_date
##   <date>
## 1 1985-04-28
## 2 1991-04-13
## 3 1992-04-18
## 4 2000-04-12
## 5 2003-04-16
## 6 2007-04-01
## 7 2009-04-16
## 8 2009-04-25
## 9 2009-05-01
## 10 2009-05-19
## # i 509 more rows
```

```
#polar_bears_no_nas %>%
#names_sep(capture_date)
```

5. (2 points) What are the min, median, mean, and max values for urea_nitrogen and creatinine?

```
polar_bears %>%
  select(urea_nitrogen,creatinine) %>%
  summarise(min_urea=min(urea_nitrogen),
            median_urea=median(urea_nitrogen),
            mean_urea=mean(urea_nitrogen),
            max_urea=max(urea_nitrogen),
            min_creatinine=min(creatinine),
            median_creatinine=median(creatinine),
            mean_creatinine=mean(creatinine),
            max_creatinine=max(creatinine))
```

```
## # A tibble: 1 × 8
##   min_urea median_urea mean_urea max_urea min_creatinine median_creatinine
##   <dbl>      <dbl>      <dbl>      <dbl>      <dbl>          <dbl>
## 1      1         11      15.1        69         0.35          1.05
## # i 2 more variables: mean_creatinine <dbl>, max_creatinine <dbl>
```

6. (3 points) As explained above, the ratio of urea_nitrogen to creatinine is used as an index of fasting. To calculate the ratio, we need to multiply urea_nitrogen by 0.466, then divide by creatinine. Create a new variable called un_creat_ratio that contains this calculation.

```
polar_bears_w_un_creat_ratio <-polar_bears %>%
  mutate(un_creat_ratio=urea_nitrogen*0.466/creatinine)
```

7. (4 points) The authors of the study categorized bears as fasting if their `un_creat_ratio` was less than or equal to 10. Use `case_when()` to create a new variable called `fasting` that categorizes bears as fasting or not.

```
#polar_bears %>%
  #mutate(fasting_polar=case_when(un_creat_ratio<=10~"fasting",
                                #un_creat_ratio>10~"non_fasting"))%>%

#select(un_creat_ratio) %>%
#mutate(fasting_polar=factor(fasting_polar,levels=c("fasting","non_fasting")))
```

```
fasting <- polar_bears_w_un_creat_ratio %>%
  filter(un_creat_ratio<=10)
```

```
non_fasting <- polar_bears_w_un_creat_ratio %>%
  filter(un_creat_ratio>10)
```

```
polar_bears_w_un_creat_ratio %>%
  summarise(fasting=un_creat_ratio<=10)
```

```
## Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
## dplyr 1.1.0.
## i Please use `reframe()` instead.
## i When switching from `summarise()` to `reframe()`, remember that `reframe()`
## always returns an ungrouped data frame and adjust accordingly.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

```
## # A tibble: 519 × 1
##   fasting
##   <lgl>
## 1 FALSE
## 2 TRUE
## 3 FALSE
## 4 TRUE
## 5 TRUE
## 6 TRUE
## 7 FALSE
## 8 FALSE
## 9 TRUE
## 10 FALSE
## # i 509 more rows
```

8. (6 points) The authors found that the number of fasting bears increased in the SB population but decreased in the CS population. Make a plot(s) that shows the number of fasting bears by year for both populations to visualize this trend.


```
#polar_bears_no_nas %>%
  #group_by(capture_date) %>%#I missed question four to have variable to put in
  #ggplot(aes(x=popn))+
  #geom_bar(fill="popn")+
  #geom_smooth()+
  #labs(title = "number of sb and cs population by years.")
```

9. (8 points) Where are the polar bears used in this study located? Make a static (not leaflet) distribution map that shows the locations of bears by population. (hint: use `color=popn` in the `aes()` of `geom_point()`)

```
polar_bears%>%
  select(lat,long) %>%
  summary()
```

```
##      lat      long
## Min.   :62.46  Min.   :-172.2
## 1st Qu.:68.11  1st Qu.:-165.2
## Median :70.53  Median :-150.7
## Mean   :69.86  Mean    :-153.7
## 3rd Qu.:71.00  3rd Qu.:-146.5
## Max.   :72.70  Max.    :-134.8
```

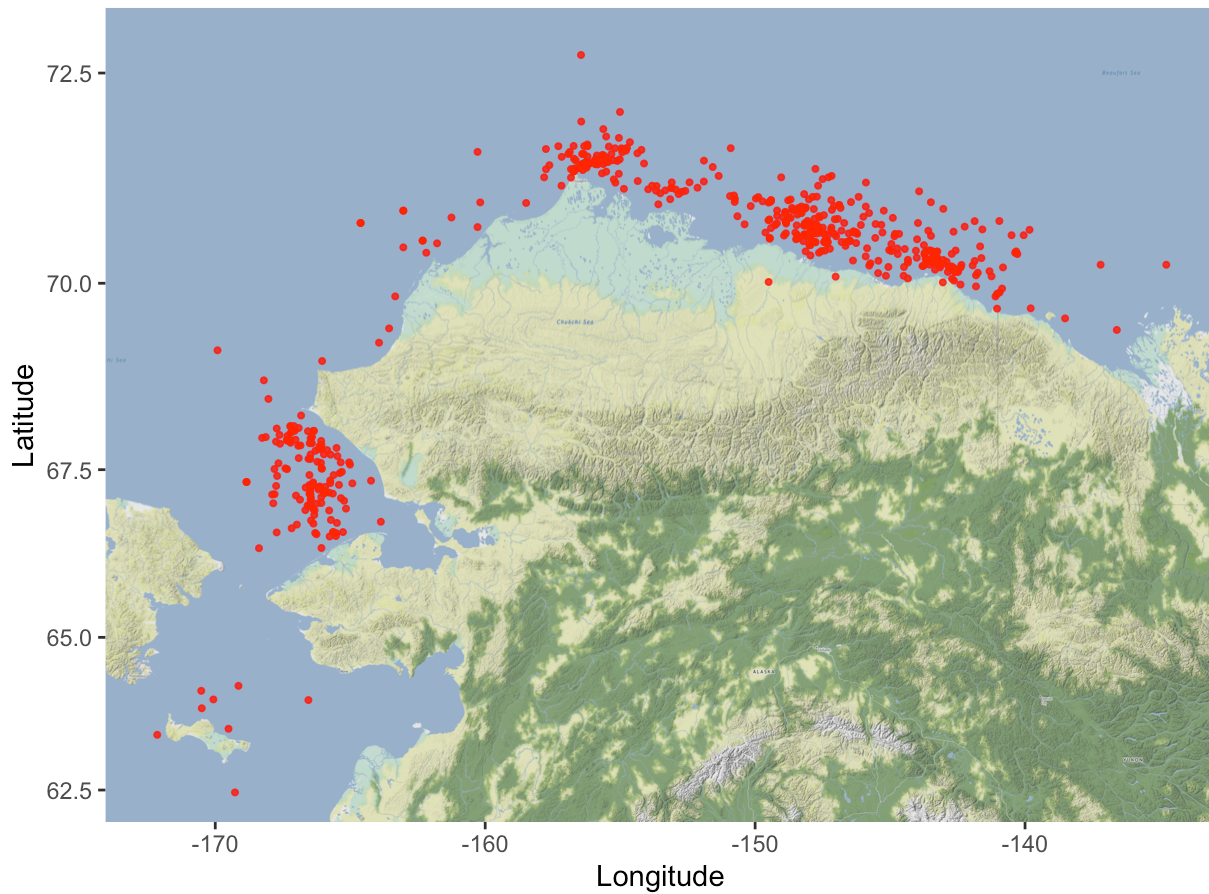
```
lat <- c(62.46,72.70)
long <- c(-134.8,-172.2)
bbox <- make_bbox(long,lat,f=0.05)
map1 <- get_stadiamap(bbox,maptype="stamen_terrain",zoom=7)
```

```
## i © Stadia Maps © Stamen Design © OpenMapTiles © OpenStreetMap contributors.
```

```
## i 165 tiles needed, this may take a while (try a smaller zoom?)
```

```
ggmap(map1) +
  geom_point(data=polar_bears,aes(long,lat),size=0.75,color="red",alpha=0.8)+
  labs(x="Longitude",y="Latitude",title="polar_bear location")
```

polar_bear location



10. (10 points) Build a simple shiny app that shows the range of `un_creat_ratio` values for both populations. The user should be able to select the fill variable and choose between sex, mating, age_class, denning, and fasting. Please use the code chunk below so that your exam knits.

```
#ui <-
  #dashboardPage(dashboardHeader(title = "Range of un_creat_ratio for different sex, mating, age_class, denning and fasting"),
    #dashboardSidebar(selectInput(inputId = "varibale",
      #label="choose variable:",
      #choices = c("Sex", "Mating", "age_class", "denning", "fasting"),
      # selected = "Sex")),
    #dashboardBody(plotOutput("plot",
      #width = "500px",
      # height = "400px"))))

#server <- function(input, output, session) {
  #output$plot <- renderPlot({
    #df <- polar_bears_w_un_creat_ratio %>%
      #filter(un_creat_ratio==input$un_creat_ratio)
    #ggplot(df, aes(x=input$variable, y=un_creat_ratio))+
      # geom_point(color=ifelse(input$un_creat_ratio=="sex", "blue", "red"))+
      # labs(title = paste("un_creat_ratio of(", input$un_creat_ratio, ")"), x="variable", y="un_creat_ratio")+
      # theme_minimal()
    # })
  #}

#shinyApp(ui, server)
```

Submit the Midterm

1. Save your work and knit the .rmd file.
2. Open the .html file and "print" it to a .pdf file in Google Chrome (not Safari).
3. Go to the class Canvas page and open Gradescope.
4. Submit your .pdf file to the midterm assignment- be sure to assign the pages to the correct questions.
5. Commit and push your work to your repository.